

STAT 24410/30030, Homework 4
Due Thursday, October 31, at 2:00pm

1. (15 pts) Lecture Notes, Chapter 3, Exercise 3.
2. (15 pts) Lecture Notes, Chapter 3, Exercise 5.
3. (15 pts) Lecture Notes, Chapter 3, Exercise 7.
4. (15 pts) Lecture Notes, Chapter 3, Exercise 8.
5. (15 pts) Lecture Notes, Chapter 3, Exercise 10.
6. (25 pts) Let $X_1, \dots, X_n$ iid $\text{Unif}(0, \theta)$. We showed in class that the method of moments estimator for θ is $\hat{\theta}_1 = 2\bar{X}_n$, and the MLE for θ is $\hat{\theta}_2 = X_{(n)} = \max(X_1, \dots, X_n)$.
 - (i) Show that $P(0 < \theta - \hat{\theta}_2 < \epsilon)$ converges to 0 as n grows to infinity. Note that this implies that the MLE is consistent. [Hint: you can start by deriving the cumulative distribution function of $X_{(n)}$.]
 - (ii) Derive the density for the MLE. [Hint: use the cumulative distribution function of $X_{(n)}$.]
 - (iii) Calculate the bias and the mean squared error for both estimators, $\hat{\theta}_1$ and $\hat{\theta}_2$.